Mind the Semantic Gap: Tool-Grounded Generation and Semantic Validation of YANG Telecom Topologies

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***Abstract*—Generating a 5G or 6G telecom topology that actually carries the service named by its intent, and not just one that passes a YANG schema check, requires checking semantics, not only syntax. A schema-valid XML instance may still attach a device to the wrong termination point, miss a bandwidth or latency budget, or satisfy syntax while failing intent-level service requirements. We address this with an Actor-Critic generator for RFC 8345-style YANG/XML telecom topologies whose critic is a tool-grounded semantic LLM validator. The same validator can also be invoked stand-alone as a commissioning auditor. The framework consolidates per-device XML descriptions into a cross-layer graph and lets an open-weight LLM choose and interpret deterministic checks for paths, bandwidth, latency, feedback requirements, and ontology compatibility. Two operating modes share the same pipeline: a diagnostic mode for audit traces and a single-round mode for faster commissioning checks. On 50 Industry 4.0 digital-twin scenarios, Qwen2.5:32B diagnostic mode reaches 92.0% accuracy and macro-F1 0.890. Qwen3:32B raises the best diagnostic result to 94.0% / 0.915. Single-round Qwen2.5 mode is 2 percentage points lower in accuracy than diagnostic mode (90.0% vs. 92.0%) and 0.052 lower in macro-F1, but cuts the median runtime from 56.5 s to 25.5 s ($\sim 2.2\times$ faster). The synthesis campaign uses the same tool-grounded critic, so generated topologies are accepted only when they pass the validator used for audit. On 500 combinatorial intents the campaign accepts 110 unique topologies (22.0%), most of them on the first or second Actor attempt.**

***Keywords*—YANG, semantic validation, telecom topology generation, large language models, tool-grounded reasoning, Actor-Critic synthesis**

I. Introduction

Automated topology generation is only useful if the generated topology can actually support the service it names. In a 5G or 6G industrial deployment, that question is not answered by XML syntax alone. A topology may conform to the IETF RFC 8345 YANG topology model and still be unusable because a camera is attached to the wrong termination point, the selected path lacks bandwidth, the latency budget is exceeded, or a feedback-required service lacks its return path. These are semantic errors: they concern the meaning of the topology with respect to an intent, not whether the XML tags are legal.

RFC 8345 gives the graph-shaped topology representation, and 3GPP TS 22.104 lists service requirements such as bandwidth, latency, and reliability targets for cyber-physical systems. Our input is a set of instantiated YANG/XML topology documents plus a structured intent. The research question is narrower and testable: can an open-weight LLM, grounded by deterministic graph tools and a telecom ontology, decide whether that topology satisfies the intent?

Raw XML is a poor interface for this decision. On campus-scale graphs, an LLM loses the relevant path among thousands of tokens and makes avoidable arithmetic mistakes. We therefore give the model a smaller, intent-focused graph and require it to use deterministic tools for the checks that should not be guessed. The paper makes three contributions. First, we build an Actor-Critic synthesis loop for YANG/XML telecom topologies in which the critic is the same semantic validator used for audit. Second, we define that validator as a three-phase pipeline: intent isolation, deterministic graph pruning, and tool-augmented LLM reasoning over RFC 8345-style instances [1]. Third, we compare diagnostic and single-round operating modes over the same validator, so the measured trade-off is the

tool-call budget rather than a different architecture.

II. Related Work

YANG and NETCONF tooling gives a strong syntactic baseline. RFC 8345 defines how network topologies are represented, and tools such as `yanglint` can check whether an XML instance conforms to its schema. They do not answer the next question: whether the graph encoded by that instance can satisfy a service intent.

Recent intent-management work uses LLMs for translation, decomposition, and assurance. INTA extracts and refines intents from network configurations [2]; other telecom studies frame LLMs as part of intent life-cycle management [3]. RAG-based approaches for 5G slices improve factual grounding [4]. These systems operate mainly on intent text, configuration text, or slice descriptions. They do not provide an executable check that a concrete RFC 8345 topology instance has a feasible path between the named endpoints, enough residual bandwidth, and acceptable latency.

Tool-augmented LLM methods such as ReAct [5] and Toolformer [6] show why external computation helps: the model can decide which operation is relevant while the tool performs the operation exactly. Telecom surveys reach a similar conclusion from the other side, reporting that direct LLM reasoning over large networks suffers from domain-adaptation errors, context degradation, and arithmetic hallucination [7]. Our work applies this pattern to topology validation: the LLM selects and interprets deterministic telecom checks, while path search, bandwidth arithmetic, latency accumulation, and ontology compatibility remain executable graph operations. The same validator then becomes a critic for synthetic topology generation.

III. System Architecture and Methodology

The framework has three components: a graph builder, a semantic validator, and an Actor-Critic generator that uses the validator as its critic. Fig. 1 sketches the architecture and the three-phase validation pipeline inside it.

A. Unified Knowledge Graph Construction

A single cross-layer industrial 5G/6G service spans several independently authored XML documents, so feeding them straight to an LLM is impractical. The framework consolidates the topology into a three-layer directed graph (device, network, application) adopting the industrial IoT connectivity layering of Szabó et al. [8], where Asset Administration Shell–based device descriptors are extended with YANG-defined connectivity attributes for visual asset management. We retain that layered representation and their port-type compatibility check, attach operational metrics to nodes and edges, and serialise the result as token-efficient JSON; the addition is a tool-grounded LLM critic (§ III.B–III.C) so the same graph supports automated semantic checking. Each physical device appears once via `supporting-node` cross-references; derived external links are added only when the inferred supporting termination points pass the validator's ontology check. Formally, $G = (V, E, \lambda, \tau, \mu_V, \mu_E)$ with $V = V_D \cup V_N \cup V_A$, $E \subseteq V \times V$, $\tau : E \to \{\text{containment, supporting, external, controller, feedback}\}$, μ_V carrying `max_bw`/`cur_bw`, and μ_E carrying `latency_ms`.

B. Deterministic Computational Offloading and Ontology

LLMs are not reliable at deterministic arithmetic or structured graph traversal [9], so the numerical and structural work is offloaded to four deterministic tools: path computation within filtered subgraphs, bandwidth feasibility against termination-point capacity, cumulative latency along discovered paths, and feedback-pair validation for QoS profiles that require a return path. Against the active QoS profile Q the tools enforce: bandwidth, $\exists P \in \mathcal{P}(s,t)$ s.t. $\forall v \in V_{bw}(P) : \mu_V(v).\mathit{max} - \mu_V(v).\mathit{cur} \geq Q.\mathit{min}$; latency, $\exists P \in \mathcal{P}(s,t)$ s.t. $\sum_{e \in P} \mu_E(e).\mathit{lat} \leq Q.\mathit{max_lat}$; and, when Q requires feedback, a matching forward–return edge pair $(a,b),(b,a)$ with compatible TPs and protocols. The path-finder enumerates up to five shortest simple paths so the latency budget passes whenever any candidate fits. Formal solvers such as Z3 or Batfish [10] could cover these graph predicates; the LLM is reserved for non-arithmetic work: mapping a natural-language intent to the right checks, arbitrating conflicting tool evidence, and producing repair hints. The ontology stores 38 node types, 21 ordered link rules, a use-case map, and 8 extensible QoS profiles in declarative tables (`HD_VIDEO` requires $\geq$ 5 Mbps, $\leq$ 150 ms, and a matching feedback link). Missing feedback edges therefore surface as deterministic failures rather than probabilistic warnings, in line with synthesis frameworks such as CEGS [11].

C. Validation Pipeline and Autonomous Synthesis

Feeding a dense network graph straight to an instruction-tuned LLM blows out the context on

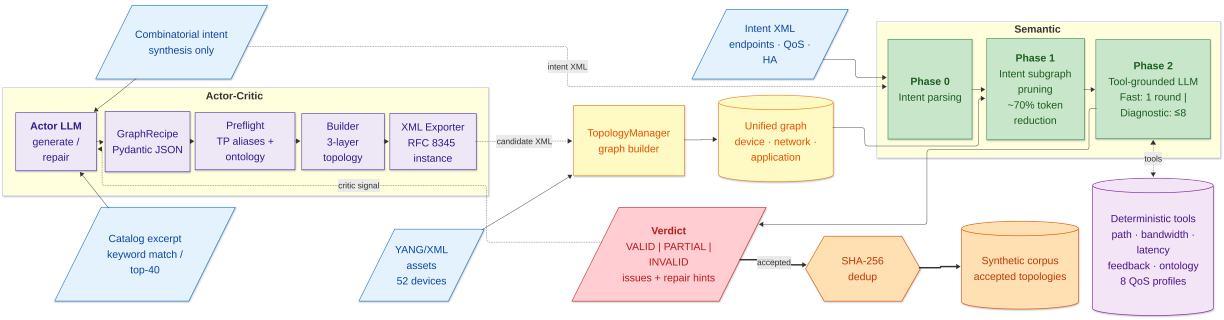


Fig. 1. Generation-and-validation framework. The validator runs three phases (intent isolation, deterministic pruning, tool-augmented reasoning) over a unified cross-layer knowledge graph; Phase 2 commits in one round (fast) or up to eight (diagnostic). The same verdict is reused as the critic of the Actor-Critic synthesis engine.

campus-scale topologies. The workflow splits into three phases. **Phase 0 (Intent Isolation)** parses the structured XML intent for endpoints, QoS profile, and high-availability flag. **Phase 1 (Deterministic Pruning)** keeps shortest-path candidates plus device hierarchies, compressing the token payload by about 70%. **Phase 2 (Tool-Augmented Reasoning)** hands the pruned subgraph to the LLM at $T{=}0.0$ with a verdict-calibration rubric. The agent must invoke each of the four deterministic tools plus the ontology checker before finalising, capped at *MAX_TOOL_ROUNDS*$=8$ and bounded by per-response (16), per-validation (40), and per-tool (6) execution limits. The pipeline runs at the network edge on an open-weight 32B model. For audit, an operator submits XML and intent and receives a verdict, evidence, warnings, and repair hints. In Fig. 2, Phases 0/1 isolate the HD-video endpoints and path; Phase 2 checks QoS, feedback, and ontology, so a schema-valid Wi-Fi-STA–to–Ethernet link is `INVALID`. The same validator also serves as critic of a two-stage synthesis loop: combinatorial intent enumeration feeds an Actor-Critic graph constructor.

Stage 1: Combinatorial Intent Enumeration. This stage uses no LLM, deliberately: the intent space is finite, and an LLM-drafted set would bias coverage toward the model's own modes. A deterministic enumerator builds tuples over 51 device profiles, 7 connectivity media, 8 QoS profiles, 7 use-case patterns, and 3 complexity tiers, filtering physically nonsensical combinations before rendering each surviving tuple into natural language. The filter is role-aware: each device profile declares a role and each QoS profile lists which roles may sit at the source and the sink, so a LiDAR reports telemetry but never streams 4K video, and feedback-required services are generated only between roles that can provide the corresponding forward and return application-layer edges. A stratified, time-seeded sampler then drew 500 intents from 11,166 valid combinations. The NL renderer only picks wording; the hard step is Stage 2, where the Actor maps an intent to a concrete topology under QoS and ontology constraints. A separate fourth tier ($4{\to}1$) is reserved as an upper-bound stress experiment.

Stage 2: Actor-Critic Graph Construction. The Actor LLM proposes a schema-enforced `GraphRecipe` given the intent and a pre-filtered catalog excerpt (intent-keyword matched, falling back to the first 40 entries) so the context does not blow out. A deterministic preflight builder rejects geometrically invalid links via the ontology module and flags reversed source/destination ordering on non-peer-to-peer connections; the validator then runs as the tool-grounded critic. `INVALID` verdicts and builder errors trigger localised repair prompts, and the loop terminates on `VALID`/`PARTIALLY_VALID` or after four attempts. The Actor uses exploration temperature $T{=}0.8$ on the first attempt and repair temperature $T{=}0.15$ on retries, under a 4096-token generation ceiling with structured truncation recovery. Accepted topologies are serialised as syntactically verified YANG/XML instances. To suppress near-identical accepted samples, each topology is canonicalised into a sorted multiset of node identifiers, layers, device types, and key attributes, then hashed with SHA-256. This signature is a practical duplicate filter for the generated corpus; it is not a proof of graph non-isomorphism. The validator-in-the-loop design [11] swaps one-shot generation for a self-correcting curation engine that runs at the edge, where errors surface as repair prompts rather than rejected outputs.

IV. Results and Performance Evaluation

A. Experimental Setup and Comparative Evaluation

The dataset is 50 Industry 4.0 digital-twin scenarios: 17 manually designed (A–P) covering basic topologies, industrial URLLC/5G-edge cases, multi-asset deployments, and campus-scale stress tests, plus 33 systematically generated (S01–S33) by injecting four fault types (bandwidth starvation, latency inflation, ontology violation, missing feedback edges) into six topology templates with one reference-VALID variant each. References are VALID/INVALID (12/38); `PARTIALLY_VALID` means hard checks pass but operator-reviewed soft risks remain, whereas missing required feedback is `INVALID`. The 76%-INVALID suite is a commissioning stress test, not evidence of broad generalisability; macro-F1 limits class-prior bias in raw accuracy. Runs use a single A100 GPU (40 GB VRAM) with Ollama serving 4-bit GGUF (Q4_K_M) at $\sim$20 GB; the 32B pipeline is launched with *num_ctx*=65536, clamped to its trained 32 k context, and the 7B baseline also runs at 32 k. Implementation, scenarios, and prompts: [12].

The proposed validator runs in two modes: **diagnostic** mode iterates up to eight tool rounds for an audit trace, while **fast** mode commits after one tool round for lower latency. Four baselines isolate one component each. **B1 (Rule-Based)** calls the same tools through deterministic `if/else`. **B2 (Tool-less)** feeds Qwen2.5:32B the raw graph with no tools, no pruning, no domain knowledge. **B3 (Prompt-augmented)** injects every QoS profile and ontology rule into the system prompt without tool access. **B4 (7B+Tools)** runs the same tool-augmented pipeline at Qwen2.5:7B and 32 k context. Table I reports the full matrix.

Diagnostic mode reaches 92.0% accuracy and macro-F1 0.890, the strongest score in the primary Qwen2.5 comparison (Qwen3 goes higher, § IV-C). Both modes catch all nine ontology faults; diagnostic recovers more borderline VALID graphs because extra rounds assemble cross-tool evidence. Even diagnostic mode still misclassifies the URLLC Wi-Fi case F (reference INVALID, predicted `VALID`), where the path is connected and ontology-compatible but cumulative latency is underweighted. The two modes share Phase 0/1; only the Phase 2 budget differs. K and S21 are reference-VALID drone fixtures missing `supporting-tp` links; their deterministic disconnects remain counted as errors.

On Scenario A (*Baseline Valid*), fast mode commits to `INVALID` after one round (8 tool calls, $\sim$25 s), while diagnostic mode's two further rounds (21 calls, $\sim$140 s) flip the verdict to the correct `VALID`; the extra rounds are what correct first-round misjudgements on borderline VALID graphs.

The rule-based B1 flags every injected fault but rejects 24/36 reference-VALID rows by treating QoS thresholds as hard cutoffs and lacking `PARTIALLY_VALID`. The tool-less B2 hedges into `PARTIALLY_VALID` on 30/150 rows (80.0% / 0.774): without an executable check, the model defaults to safe warnings. Prompt-augmented B3 isolates that effect cleanly – same model with every QoS profile and ontology rule in-prompt but no tools collapses `VALID` recall to 0.083 (macro-F1 0.557). Domain text alone tilts the model toward hedging; tool grounding flips it back to commitment. B4 (7B+tools) reaches 84.0% / 0.729; tool augmentation helps below 32B on accuracy but `VALID` recall stays low, marking a reasoning-capacity floor.

B. Ablation Study and Scalability

The two structural ablations in Table I (lower block) each remove one component. $-$**Pruning** ties rule-based at 84.0% accuracy with `VALID` recall 0.333: feeding the full topology graph to Phase 2 dilutes attention on the VALID scenarios, although it does not break the validator structurally and completes within the tool-round budget. $-$**Ontology tools** drops to 82.0% with macro-F1 0.781 and a flipped failure profile (`VALID` recall 0.833, `INVALID` recall 0.816): without ontology grounding the model accepts off-path violations as benign on the `INVALID` side and over-commits to `VALID` on borderline VALID scenarios. Pruning also compresses the token payload by about 70%, the single largest contributor to the diagnostic mode's tractability on campus-scale graphs.

C. Model-Family Generalisation

Two additional models ran both modes on the 50-scenario suite with everything else held constant: Qwen3:32B (next-generation Qwen) and Gemma4:31B (a different family, same quantisation). Diagnostic mode transfers across both without architecture changes, and Qwen3 is the strongest configuration in the matrix at 94.0% / 0.915. Almost all of that gain is in the `VALID` class, where Qwen3 lifts recall from 0.750 to 0.833 and F1 from 0.818 to 0.870 over Qwen2.5 while `INVALID`-class metrics barely move; the borderline topologies Qwen2.5 hedges into `PARTIALLY_VALID` are the ones Qwen3 commits to `VALID` on the same evidence.

TABLE I

COMPARATIVE EVALUATION ON 50 INDUSTRY 4.0 SCENARIOS. SCENARIO-LEVEL MAJORITY VERDICTS; 3 ITERATIONS FOR DIAGNOSTIC MODE AND BASELINES B1/B2 (n=150), 1 ITERATION FOR FAST MODE, B4, ABLATIONS, AND THE MODEL-GENERALISATION ROWS (n=50). MACRO-F1 AVERAGES THE PER-CLASS F1 OF `VALID` AND `INVALID` ONE-VS-REST.

			VALID			INVALID		
Method	**Acc.**	**$F1_M$**	P	R	F1	P	R	F1
Diagnostic mode (Qwen2.5:32B, multi-turn)	**92.0%**	**0.890**	.900	**.750**	**.818**	.949	.974	**.961**
Fast mode (Qwen2.5:32B, 1 round)	90.0%	0.838	**1.000**	.583	.737	.884	**1.000**	.938
B1: Rule-based	84.0%	.702	**1.000**	.333	.500	.826	**1.000**	.905
B2: Tool-less (Qwen2.5:32B)	80.0%	.774	**1.000**	.417	.588	**1.000**	.921	.959
B3: Prompt-augmented (Qwen2.5:32B)	74.0%	.557	**1.000**	.083	.154	.973	.947	.960
B4: 7B + tools (Qwen2.5:7B)	84.0%	.729	.833	.417	.556	.841	.974	.902
Ablations and model-family generalisation (1 iteration, n=50)								
−Pruning	84.0%	.702	**1.000**	.333	.500	.826	**1.000**	.905
−Ontology tools	82.0%	.781	.588	.833	.690	.939	.816	.873
Diagnostic mode (Qwen3:32B)	**94.0%**	**0.915**	.909	.833	.870	.949	.974	.961
Fast mode (Qwen3:32B)	76.0%	.432	.000	.000	.000	.760	**1.000**	.864
Diagnostic mode (Gemma4:31B)	80.0%	.681	**1.000**	.250	.400	.949	.974	.961
Fast mode (Gemma4:31B)	76.0%	.432	.000	.000	.000	.760	**1.000**	.864

Gemma4 transfers only partially (80.0% / 0.681, `VALID` recall 0.250), so the in-prompt verdict rubric is evidently calibrated for Qwen-family decoding.

Fast mode is far more model-sensitive. On both Qwen3 and Gemma4 it collapses to a flat all-`INVALID` classifier (`VALID` F1 zero; the 76.0% accuracy is the dataset prior alone), and both models default to `INVALID` even on easy scenarios like `A`. Single-pass commitment needs a model already calibrated to commit after one tool pass, which Qwen2.5:32B is and the other two are not. Thus diagnostic is the default across uncalibrated models; fast mode requires model-specific validation.

D. Autonomous Topology Synthesis Evaluation

The Actor-Critic synthesis subsystem is evaluated on a scaled campaign of 500 combinatorial intents sampled from the enumeration described in § III.C, with role-aware source/sink and device–medium affinity constraints applied before emission. Up to four repair attempts are allowed per intent, with Actor temperature 0.8 on the first attempt and 0.15 on retries. The campaign is run with the tool-grounded critic: an accepted topology must clear the same Phase 0/1/2 validator used in the evaluation suite, not only a deterministic preflight. The campaign accepts 110 of 500 intents (22.0%) with 0 duplicate signatures, and the accepted graphs span 13–30 nodes and 14–40 edges (mean 20.7/23.8). Acceptances cluster on the early attempts: 52 land on attempt 1, 45 on attempt 2, 11 on attempt 3, and 2 on attempt 4, so the four-attempt repair budget is rarely exhausted on a topology that ever clears.

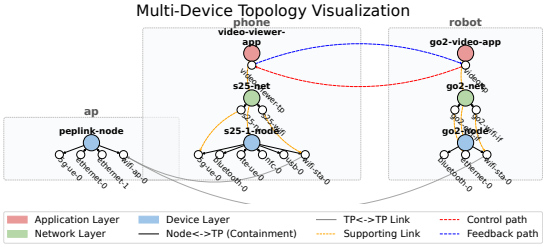


Fig. 2. Generated topology visualization with three device groups labelled `ap`, `phone`, and `robot`. Red, green, and blue nodes denote the application, network, and device layers shown in the legend; black edges connect nodes to their termination points, gray edges connect termination points, and orange dashed edges denote supporting links. The Peplink node exposes `wifi-ap-0`, which is linked to Wi-Fi station termination points in the phone and robot groups. At the application layer, `go2-video-app`/`video-tp` and `video-viewer-app`/`video-viewer-tp` are connected by a red dashed forward service path from robot to phone and a blue dashed return feedback path from phone to robot; the validator treats the pair as a service-specific feedback requirement within semantic validation.

Repair uses the same feedback channel for validator and builder failures: JSON or Pydantic parse errors become structured `builder_errors` sent back to the Actor on the next attempt instead of aborting the trial at attempt 0. Fig. 2 shows one accepted topology drawn from the campaign output; the layered structure renders one column per physical device with three stacked horizontal layers and ontology-checked derived links, as described in § III.A.

The deterministic preflight rejects geometries with orphan devices before the validator critic ever runs, so every accepted topology must have every selected asset

participating in at least one external link. The catalog auto-includes a Wi-Fi/Ethernet bridge for Wi-Fi device classes. The rejection breakdown explains the 22% curation yield: bandwidth (≈39%), ontology/builder violations (≈38%), duplicates (≈9%), missing paths (≈7%), and parse aborts (≈5%); 97/110 acceptances clear by attempt 2.

V. Discussion

Because both modes share Phases 0/1, the choice is operational: diagnostic better recovers borderline positives across model families, while fast halves latency only on calibrated Qwen2.5. On synthesis, the useful split is between builder-side errors, validator-side semantic failures, and duplicate accepted graphs. SHA-256 deduplication is deliberately conservative: it keeps repeated Actor samples out of the corpus, but it does not claim to measure full structural graph diversity.

a) Limitations and scope: The 50-scenario suite is 76% INVALID and has only 12 positives, so it tests fault-heavy commissioning rather than broad generalisability. K/S21 omit `supporting-tp` links; scores retain their intended-VALID labels, while future balanced suites should correct the fixtures and add operator cases. The validator currently focuses on pairwise TP compatibility, path QoS, and feedback-edge presence for service profiles that require it, and does not yet cover richer service composition or isolation patterns. For generation, the Actor may also switch a requested Wi-Fi or 5G source to Ethernet if that yields a simpler graph the critic still accepts.

VI. Conclusion and Future Work

Telecom YANG topologies need semantic validation beyond syntactic checks, especially when generated automatically. We presented an Actor-Critic generation loop whose critic is a tool-grounded LLM validator, evaluated in diagnostic and single-round modes. Multi-turn reasoning is slower but much less brittle across model families, and the same semantic machinery audits topologies and curates synthetic ones when the critic runs in the same tool-augmented mode. The 500-intent campaign confirms the loop end-to-end, accepting 110 unique topologies (mean 21 nodes, 24 edges).

Future hybrid routing could send only prechecked, calibrated low-risk cases to fast mode and all borderline or ontology-suspicious cases to diagnostic mode; it is not evaluated here. The validated RFC 8345 graphs also yield labelled supervision for downstream GNN/GAN topology learning, though deployment-grade synthesis still needs human review.

Acknowledgment

This work has been part of Celtic-Next project RAI-6Green, it was supported by project 2024-1.2.6-EUREKA-2024-00009 financed under the 2024-1.2.6-EUREKA scheme, implemented with support provided by the Ministry of Culture and Innovation of Hungary from the National Research, Development and Innovation Fund.

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